Anik Chakraborty

M.Sc. Economics (1st Year)

Indian Institute of Technology, Kanpur

anikc25@iitk.ac.in | +919002398090

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2025 - Present	M.Sc Economics	Indian Institute of Technology, Kanpur	10/10
2025	B.S.(Hons) Economics	Indian Institute of Technology, Kharagpur	8.54/10
2021	Class XII (WBCHSE)	Ghatal Vidyasagar High School	98.00%
2019	Class X (WBBSE)	Ghatal Vidyasagar High School	97.71%

Scholastic Achievements

- Achieved an **All India Rank of 22** among 2,973 candidates who appeared in the **GATE 2025 Economics** examination.
- Secured an **All India Rank of 4,474** among more than 1.41 lakh candidates who took the **JEE Advanced 2021** exam.
- Attained an **All India Rank of 2,907** among 10 lakh candidates who appeared for the **JEE Main 2021** exam nationwide.
- Secured **55th** rank among 65 thousand candidates appeared in the **West Bengal Joint Entrance Examination** in **2021**.
- Ranked **73rd** at the national level in the **ISI B.Stat 2021** entrance examination conducted by the Indian Statistical Institute.
- Secured **State Rank 2** in the **National Talent Search Examination Stage-I 2019** held at the state level in West Bengal.
- Awarded the **NTSE** scholarship after clearing the Stage-II examination in **2019**, conducted by NCERT at the national level.

Key Projects

Role of dividend Policies in Stock Valuation | BTP, under Dr. Gourishankar S Hiremath, IIT Kharagpur *Jan'25-Apr'25*

- Investigated how different **dividend policies** affect **stock valuation** by analyzing **S&P 500 companies** from **January 2015** to **December 2024**, covering both dividend-paying firms and non-dividend-paying firms across various industry sectors.
- Used **event study methodology** to capture immediate **market reactions** to **dividend announcements** and applied **panel regression** and **machine learning models** (**Random Forest, GBM, SVR**) for long-term performance analysis.
- Found that **dividend increases** led to **positive abnormal returns**, while **cuts** triggered **negative market reactions**; metrics like **dividend yield**, **payout ratio** predicted **risk-adjusted returns** with model **R-square** value exceeding **0.70**.
- Concluded that firms with **stable, progressive dividend policies** deliver **better risk-adjusted returns** and attract **stable, long-term investors**, underscoring the **strategic value** of consistent and effective dividend management practices.

Stock Price Prediction:Tata Motors | BTP, under Dr. Gourishankar S Hiremath, IIT Kharagpur *Jul'24-Nov'24*

- **Forecasted Tata Motors**' daily closing stock price using **Linear Regression**, **Moving Averages**, **Random Forest**, and **Long Short-Term Memory (LSTM)** networks, trained on comprehensive historical stock market data from **2018–2024**.
- **Engineered** multiple time-series features including **30/100-day moving averages**, **volume variability**, and applied **Min-Max scaling** with a **60-day sequence window** to optimize data input for effective LSTM model training and performance.
- **Evaluated** various model performances using **Root Mean Squared Error(RMSE)** and **Mean Absolute Percentage Error(MAPE)**; **LSTM achieved lowest RMSE (30.06)** and **MAPE (2.44%)**, outperforming all other models tested.
- **Demonstrated** that **LSTM networks** effectively captured complex and non-linear time-series dependencies, offering strong, consistent, and reliable performance for **short-term stock price prediction** in highly volatile financial market conditions.

Valuation of Adidas AG Using Direct Cash Flow Models | Course Project, IIT Kharagpur *Apr'24*

- Conducted a comprehensive **equity valuation** project aimed at thoroughly analyzing **Adidas AG**'s financial structure, performance metrics, market standing to estimate its **true enterprise value** and assess investment potential to the stock.
- Performed an in-depth **ratio analysis** across liquidity, activity, profitability, and solvency metrics, and built a comprehensive **5-year financial forecast** incorporating detailed growth projections, margin assumptions, and operational efficiency trends.
- Applied three distinct valuation techniques such as **FCFF**, **CCF**, and **ECF** and used a calculated **WACC (8.45%)** in conjunction with **sensitivity analysis** to model various economic and business case scenarios for robust enterprise valuation.
- Found that **Adidas** was significantly **undervalued** by approximately **20%**, with its **intrinsic enterprise value** estimated between **€45B–€46B**, compared to the actual valuation of **€37.5B** in the fiscal year **2023**, based on all valuation models.

Technical Skills

- **Programming Languages:** Python, C Programming
- **Software and Libraries :** Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, LaTeX, MySQL, Stata, MS Excel

Relevant Courses

Probability and Statistics	Statistics for Economics	Indian Economy
Derivatives	Econometric Analysis	Economic Data Analysis
Financial Management	Corporate Valuation and Restructuring	Equity Research
AI & ML	Linear Algebra, Numerical and Complex Analysis	Advanced Calculus
Microeconomics	Financial Institutions and Markets	Macroeconomics
Development Economics	Programming and Data Structures	Public Finance
Economics of Growth	General Equilibrium and Welfare Analysis	International Economics

Online Course and Certification

- Python for Data Science and Machine Learning Bootcamp, Udemy. (*Aug, 2025*)